
ELK336-Quantum Computation

HOMEWORK

Due 31.03.2023

- The students are obligated to do the homework themselves.
 - The copied homework will not be considered or graded.
 - Homework should be ready before 23.00 on the ninova in electronic form.
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- 1- Assume that $\{|a\rangle, |b\rangle, |c\rangle\}$ forms an orthonormal basis in $\mathbb{C}^3$ and $|\psi\rangle = \frac{1}{\sqrt{5}}|a\rangle - i\sqrt{\frac{7}{15}}|b\rangle + \frac{1}{\sqrt{3}}|c\rangle$ is given.
Calculate the inner product $\langle\psi|\psi\rangle = ?$ Is $|\psi\rangle$ state normalized?
 - 2- Given Pauli matrices X, Y, Z ; show that $X^2 = Y^2 = Z^2 = \mathbb{1}$
 - (a) by using the matrix representation
 - (b) by using outer product notation.
 - 3- Given $A = i|1\rangle\langle 1| + \frac{\sqrt{3}}{2}|1\rangle\langle 2| + 2|2\rangle\langle 1| - |2\rangle\langle 3|$ with basis $\{|1\rangle, |2\rangle, |3\rangle\}$ in $\mathbb{C}^3$,
 - (a) Find $A^\dagger$.
 - (b) Show if A is unitary or not.
 - 4- (a) Show that $XYX = -Y$ and (b) show that $[XY, YZ] = -2iY$.
 - 5- It is known that computational basis states are $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.
Consider a vector state $|\psi\rangle = |10\rangle$ which is one two-qubit basis states.
Assume an operator A which applies the Pauli- X operation only to the second qubit.
 - (a) Show that the $A|\psi\rangle = |11\rangle$ by using matrix representation.
 - (b) Obtain the same result by Dirac notation.
 - 6- (a) Given $K = A^\dagger B \otimes CD^\dagger$ and $L = A^\dagger \otimes C$, find the simplest form of $K^\dagger L = ?$
(b) Find the matrix representations of $|0\rangle\langle 0| \otimes Z + |1\rangle\langle 1| \otimes \mathbb{1}$ and $|0\rangle\langle 0| \otimes \mathbb{1} + |1\rangle\langle 1| \otimes Z$, which quantum gates are they? Draw their circuit representations.
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The questions will be evaluated as 1- 15 pts., 2-(a) 5pts., 2-(b) 10pts., 3- (a) 5 pts., 3-(b) 10 pts. 4-(a) 5 pts, 4-(b) 10 pts 5-(a) 10pts, 5-(b) 5 pts. 6-(a) 10pts 6-(b) 15pts